Quark State Visualization in the Probabilistic-Origin Framework

Spin ⊗ Flavor ⊗ Color ⊗ Origin (±0)

This document explains how quark properties (spin, flavor, color) can be visualized when nested inside the probabilistic-origin ±0 framework.

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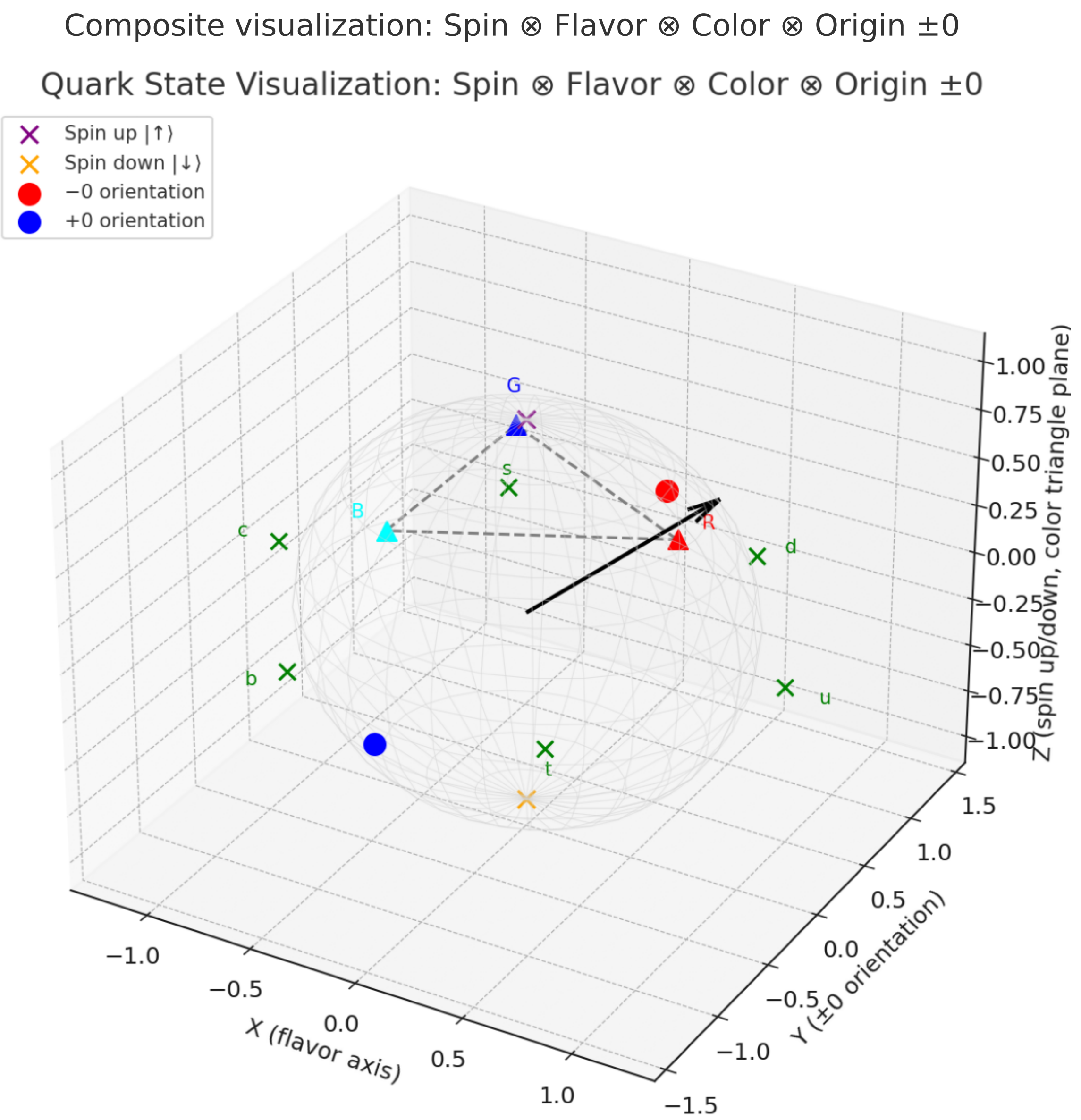
Project: Neural Optical Co-Processor


Composite visualization: Spin ⊗ Flavor ⊗ Color ⊗ Origin ±0
Quark State Visualization: Spin ⊗ Flavor ⊗ Color ⊗ Origin ±0
Spin up |↑⟩
Spin down |↓⟩
−0 orientation
+0 orientation
G
B
R
s
c
d
b
u
t
X (flavor axis)
Y (±0 orientation)
Z (spin up/down, color triangle plane)
−1.0
−0.5
0.0
0.5
1.0
1.5
−1.5
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

Breakdown of Layers

- **Spin (Bloch sphere):**
  Quarks are spin-½ particles. The purple/orange poles mark |↑> and |↓>.
  Superpositions appear as vectors on the sphere.

- **Flavor (hexagon):**
  The six quark flavors (u, d, s, c, b, t) are represented as a polygon
  around the equator plane. Flavor oscillations can be seen as rotations
  around this hexagon.

- **Color (SU(3) triangle):**
  Quarks carry a color charge (R, G, B). Here shown as a triangular set of
  states above the equator plane, connected by dashed edges.

- **Origin orientation (±0):**
  The probabilistic origin adds an extra qubit (red = −0, blue = +0).
  Collapse aligns spin, flavor, and color relative to one of these orientations,
  which then converge to an effective 0 for lab measurement.

Collapse Picture

- Measurement collapses spin (up/down), flavor (one of six), color (R/G/B),
  and orientation (±0) **simultaneously**.
- Orientation collapse ensures that −0/+0 are indistinguishable in the lab frame,
  but they matter for how superpositions and entanglement distribute.

Implications

- The framework suggests quarks are not just spin⊗flavor⊗color, but also include
  a reference qubit (origin ±0) that could play a role in how correlations and
  entanglement are observed.
- This could offer new ways to visualize flavor oscillations, spin correlations,
  and possibly subtle asymmetries in high-energy experiments.